

Jack Cerrone

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Education

Charles W. Baker High School | Class of 2026

- 97.75 Unweighted GPA | 99.94 Weighted GPA

Onondaga Community College | Non Matriculated Student

- Summer coursework in advanced mathematics | 4.0 College GPA

Rensselaer Polytechnic Institute | Class of 2030 | Majoring in Mathematics and Physics

- Awarded the Rensselaer Medal for excellence in mathematics and science

New Visions Engineering at Lockheed Martin

- Part of an elite collective of motivated students based at Lockheed Martin who are interested in STEM fields

Skills

Mathematics

- Calculus (Differential, Integral, and Multivariable): Differentiation and integration techniques, multivariable and vector calculus, optimization, and analytical problem solving
- Differential Equations (Ordinary and Partial): Analytical/numerical solution methods, math modeling
- Linear Algebra: Matrix operations, vector spaces, eigenvalues/eigenvectors
- Advanced Mathematics (Tensor Calculus): Coordinate transformations and basics of general relativity

Technical Skills

- Programming: Python, C++, MATLAB - scripting and mathematical computation
- CAD: Autodesk Inventor (4 years) - part design, assemblies, and technical drawings

Other

- Analytical Problem Solving: Applying mathematical reasoning to complex physics and engineering problems
- Technical Documentation: Writing technical reports, procedures, and documentation using Microsoft Suite

Work Experience

Lockheed Martin | Systems Engineering and Computational Analysis Intern (2025 - 2026)

- Used MATLAB to generate probabilistic detection maps for radar systems under numerous operational scenarios to verify system requirements
- Supported the documentation and computational analysis of the TPS-77, TPS-77 MRR, and FPS-117 systems as well as assisting with the integration, verification, and validation of system requirements across multiple subsystems
- Performed radar test procedures to verify system requirements outside of the computational realm

Leadership & Extracurricular Activities

Founder and President of Calculus Club (2024 - 2026)

- Established an after school club dedicated to helping students understand new calculus concepts
- Designed lesson plans and tutoring sessions that improved members' comprehension in higher math
- Fostered a collaborative learning environment encouraging student participation and growth

Mathematics and Physics Tutoring (2024 - 2026)

- Provide group and individual tutoring for peers struggling with mathematics and physics, helping improve grades and understanding of key concepts
- Provided specialized tutoring for around 30 students for a combined total of roughly 165 hours
- Adapt teaching methods to fit individual learning styles for maximum impact

Founder and President of Engineering Team (2024 - 2026)

- Competed in the Real World Design Challenge (RWDC) as the Team and Design Lead for a 7 member aerospace engineering team
- Designed an industrial hexacopter for the delivery of critical supplies to first responders during wildfire operations
- Presented final solution to industry engineers at Lockheed Martin under Chief Engineer Jeff Marier
- First place among competing teams across New York State and invited to present in an international conference